Blockchain Technology (CSE4444)

Assignment 2: Decentralized Application (DApp) Submission

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Assignment Title:	Blockchain Technology (CSE4444) - Assignment 2
Title of Developed DApp:	Blockchain-Based Student Attendance Management System
GitHub Repository Link:	https://github.com/Sakshi-M16/blockchain-student-attendance

1. Problem Statement & System Overview

Traditional educational attendance tracking relies on manual paperwork or centralized databases vulnerable to retroactive modifications, proxy logging, and lack of transparency. The **Blockchain-Based Student Attendance Management System** is an original decentralized application built on Ethereum. Using smart contracts, attendance logs are secured immutably, allowing only the authorized instructor to record attendance while enabling students to verify their genuine records transparently.

2. Smart Contract Architecture

The smart contract is written in Solidity (v0.8.20) in contracts/AttendanceSystem.sol. Key features include:
• **Access Control:** onlyInstructor modifier restricts registration and attendance logging strictly to the deployer.
• **Validation:** Reverts on zero roll numbers, empty strings, duplicate registrations, and duplicate attendance dates.
• **Events:** Emits StudentRegistered, AttendanceMarked, and LectureConducted.
• **View Functions:** getStudent, isAttendedOnDate, getAttendancePercentage, and getAllStudents.

3. Frontend & MetaMask Integration

A responsive user interface was created with HTML5, CSS3, and JavaScript, integrated with Ethers.js v5.7.2. MetaMask is used for account authentication and transaction signing. The DApp dynamically detects network changes, verifies the instructor role, submits write transactions to the blockchain, and queries live ledger data without gas cost.

4. Comprehensive Testing & Results

TC	Category	Scenario / Revert Condition	Observed Result
TC-01	Valid	Deploy with courseName parameter	**PASS** - Deployed
TC-02	Valid	Register student (101, 'Alice Johnson')	**PASS** - Registered
TC-03	Valid	Mark attendance (101, 20261017, Present)	**PASS** - Recorded
TC-04	Valid	Query student record and attendance %	**PASS** - Displayed
TC-05	Invalid	Register student with empty name ('')	**PASS** - Reverted
TC-06	Invalid	Register student with roll number 0	**PASS** - Reverted

TC-07	Invalid	Duplicate student registration (101)	**PASS** - Reverted
TC-08	Invalid	Attendance for unregistered student (999)	**PASS** - Reverted
TC-09	Invalid	Duplicate attendance on same date	**PASS** - Reverted
TC-10	Unauthorized	Non-instructor calls registerStudent	**PASS** - Reverted
TC-11	Unauthorized	Non-instructor calls markAttendance	**PASS** - Reverted

5. Submission Checklist & Repository Organization

• contracts/AttendanceSystem.sol: Solidity Smart Contract source code.
• frontend/: index.html, style.css, app.js, and contract_abi.js.
• deployment/deployment_details.md: Remix IDE, Ganache, and Sepolia deployment steps.
• test/test_contract.py & test_results.md: Complete test suite and test results.
• screenshots/: Compilation, deployment, MetaMask, and transaction verification evidence.
• README.md: Project execution, setup guide, and contract interaction documentation.